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Heat and moisture transfer in fabric consisting of fibers with low hygroscopicity

Yang Yang[1,2], Xungai Wang[3], Zheng Liu[1,2], Chenhua Zheng[1,2], Lin Lou[4,5], Xin Liu[3] and Peihua Zhang[6]

Abstract

The heat and moisture transfer through dry and wet fabric made up of hydrophobic fibers was modeled and experimentally verified. In the modeling, the governing equations are obtained with consideration of the energy balance and the mass balance of the liquid water, where the specific initial and boundary conditions were set according to measurement of the cooling property of fabric. By comparing temperature changes at the fabric inner surface, the model validity was confirmed by the agreements between the experimental results and numerical solutions. Thus, this simulation was able to predict the cooling performance of fabrics. The effects of internal and external factors on heating/cooling performances of fabrics were analyzed by this model. With an increasing evaporation rate, thermal conductivity and water content of fabric, the cooling capability of fabric was improved. Surrounding conditions with lower temperature, lower relative humidity and higher air velocity demonstrated a positive effect on fabric thermal and moisture transfer performances, which benefits the thermal comfort of the human body after excessive sweating.

Keywords

Mathematical model, heat and moisture transfer, heating/cooling performance, influence factors

Heat transfer plays an important role in clothing thermal comfort.[1,2] Fabrics of summer or sports clothing are required to have excellent thermal and moisture transfer capabilities to keep the microclimate of the body cool, dry and comfortable. The cooling property of fabric can be evaluated through a process of heating dry and wet fabric.[3] The heating performance of dry fabric shows the fabric heat transfer and radiation, whereas the heating property of wet fabric demonstrates the combined effects of coupled heat–moisture performance, and the cooling function of moisture evaporation and diffusion of wet fabric. With decreasing fabric temperature at the same heating time, the cooling efficacy of fabric is enhanced.

Thermal comfort is highly dependent on the heat and moisture transfer performance of clothing.[2,4] Heat can be transferred faster to the external ambient environment through fabric with good heat conduction, which is beneficial for cooling the human body. The heat and moisture transfer behaviors are affected by the activity level of the human body[5] and the surrounding conditions, such as temperature, relative air velocity and vapor pressure in ambient air. Lower temperature and vapor pressure and higher air velocity

[1]Clothing Engineering Research Center of Zhejiang Province, Zhejiang Sci-Tech University, China
[2]School of Fashion Design & Engineering, Zhejiang Sci-Tech University, China
[3]Deakin University, Institute for Frontier Materials, Australia
[4]Key Laboratory of Advanced Textile Materials and Manufacturing Technology (Zhejiang Sci-Tech University), Ministry of Education, China
[5]Zhejiang Provincial Key Laboratory of Fiber Materials and Manufacturing Technology, Zhejiang Sci-Tech University, China
[6]College of Textiles, Donghua University, China

Corresponding authors:
Xin Liu, Deakin University, 75 Pigdons Road, Waurn Ponds, Victoria 3216, Australia.
Email: xin.liu@deakin.edu.au

Peihua Zhang, Donghua University, 2999 Renmin Road (North), Songjiang District, Shanghai 201620, China.
Email: phzh@dhu.edu.cn

promote heat and moisture transfer when a body sweats excessively. In addition, it is known that at low activity levels, there is no liquid sweat produced by the human body, and the heat transfers of clothing are mainly determined by heat conduction, convection and radiation.[6] This energy exchange of dry heat is not effective enough for releasing heat from the human body at high activity levels or in a hot environment, and excessive heat leads to poor thermal comfort and low work efficiency.[7] After sweating profusely, moisture evaporation becomes central to maintaining the human body at a comfortable level.[8] This is because excellent moisture evaporation takes away a large amount of heat produced by the cooling sensation.[9–13] Thus, during exercise or in hot conditions, the thermal sensation of a clothed body is mainly affected by the sweating rather than the heat transfer performance of fabric due to the excellent cooling effect triggered by sweat evaporation.[14] There are three main mechanisms for evaporative heat loss: water diffusion, sweat secretion and water evaporation.[15] To predict the sweat behavior, a multi-model consisting of a rational model and an empirical model has been proposed.[16] This model could be effectively used to predict the average sweat loss and the sweating rate, respectively.

Various simulations have been carried out to investigate the coupled heat and moisture transfer behavior in porous media. Based on the assumption that moisture content is linearly related to temperature, Henry[17] proposed a mathematical model with consideration of the moisture sorption/desorption of wool fibers to investigate the transfer behaviors of coupled heat and moisture in wool textiles. Henry's model was improved later by David and Nordon,[18] and the assumptions used in the model were more consistent with the actual moisture sorption process. This model provided a space–time relationship between moisture concentration and temperature, whereas the fiber sorption mechanisms in the model were ignored and it required an experimental coefficient of the sorption rate. To optimize this model, Li and Holcombe[19] developed a two-stage model considering the moisture sorption mechanisms to analyze the relationship between the water vapor absorption rate of fibers and the coupled heat and moisture transport in a wool fabric. Furthermore, the effect of dynamic heat and moisture transfer behavior of clothing on the human thermoregulation system was described by modification of Gagge's two-node model and coupled heat and moisture transfer equations.[20] The perception of coolness and dampness sensation also could be predicted through a mathematical simulation that describes the heat and moisture transfer mechanisms in fabrics and the neurophysiological responses of thermoreceptors.[21] Fontana et al.[22] numerically simulated heat and moisture transfer in a thermal insulation fabric by the finite volume method in a cold environment. Similarly, Xu et al.[23] proposed a numerical model to describe the dynamic behavior of heat and moisture transfer of porous fabric in an environment with low temperature. Their models were employed to compute the temperature distributions, moisture concentration and water content in the fabric. The heat and moisture transfer during the high heat flux condition based on the drying process[24] and heat transmission through a fabric during the measurement of thermal resistance[25] was also simulated. It is noticed that the above works are mainly focused on the heat and moisture transfer of fabric with low moisture content (moisture regain in the test environment), and one of the most important parts is simulation of the moisture absorption of fibers. There are strong agreements between their predictions and experiments for heat and moisture transfer in porous fabric. However, the thermal and moisture transfer behaviors of fabrics with a high moisture content in the cooling process have not been investigated deeply. Therefore, compared to using fabric with a low moisture content to study the thermal and moisture transfer, using fabric with a high moisture content to investigate the critical role of evaporative heat losses on a cooling body is highly desirable.

Regarding thermal-wet comfort fabrics, fibers with low hygroscopicity have been widely used because hydrophobic material is conducive to water transfer.[26,27] In particular, polyester and nylon have a broad application in the fabrication of comfort fabrics due to their excellent physical and chemical characteristics.[28,29] Polyester fibers demonstrate low hygroscopic performance (official moisture regain, 0.4%),[30] and nylon fibers have limited moisture absorption performance (official moisture regain, 4.5%). Thus, the simulation of thermal and moisture transfer in the evaporation process will benefit the investigation of the cooling mechanism of fabric made up of fibers with low hygroscopicity.

In our previous work, an evaluation method for the cooling properties of fabrics was proposed.[3] The thermal performances of dry and wet fabrics were tested respectively to estimate the cooling function of fabrics. In this work, to better understand the cooling mechanism of fabric, a mathematical model was developed that provides an insight into heat and moisture transfer, evaluating the cooling properties. The computational and experimental temperature distributions were compared. Meanwhile, the influences of the thermal conductivity and the evaporation rate of fabric, the water content within fabric and the surrounding conditions (temperature, relative humidity and air velocity) on the heat and moisture transfer behavior were investigated through this simulation. The results could be

used to evaluate the cooling performances of polyester fabric more simply and effectively and provide informative data for the garment manufacturing industry to develop new applications for polyester fabric.

Model construction

Model assumptions

The model is established on the basis of differential heat transfer and combined heat and mass transfer in dry and wet fabrics. In this work, the fabric is regarded as a stationary and homogeneous porous solid. The principal model assumptions are listed as below.

(1) Water sorption/desorption by the porous fabric was neglected due to the low moisture regain and poor hydrophilicity of polyester fabric.
(2) The fiber volume changes due to moisture absorption are neglected.
(3) A local thermal equilibrium exists among all phases (solid, liquid and gas). All phases at the same location have the same temperature.
(4) The heat radiation in the fabric is neglected.
(5) Water vapor flow is unconstrained by the pores of the fabric, so the water vapor transfer from fabric to the surrounding space is not a rate-limiting process.
(6) A kinetic model is used to describe the liquid evaporation. It assumes that the liquid evaporation is proportional to a gap between the equilibrium and the actual local water vapor pressure (see Equation (4)).
(7) The liquid only diffuses along the thickness direction of the fabric.

Geometry

The testing process of the fabric cooling property is simple. The fabric sample was placed horizontally on the heating board with settled power. A two-dimensional (2D) model was constructed according to the testing process. The width of the heating board is 200 mm, while the height is 5 mm. The width of the fabric is the same as that of the heating board, while the height is the fabric thickness.

Governing equations

The liquid-phase mass balance is established in Equation (1). The energy balance is considered in Equation (2). The governing equations are listed as follows

$$\frac{\partial \varepsilon_l}{\partial t} = \frac{\partial}{\partial x}\left(D_l \frac{\partial \varepsilon_l}{\partial x}\right) - \frac{\dot{m}_{evp}}{\rho_l} \tag{1}$$

$$\rho_{com} c_{v,com} \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(\lambda_{com} \frac{\partial T}{\partial x}\right) + Q \tag{2}$$

$$\varepsilon_f + \varepsilon_a + \varepsilon_l = 1 \tag{3}$$

where ε_l, ε_f and ε_a represent the volume fractions of liquid water, fibers and air, respectively; t is time, s; x is the location along the thickness of fabric, mm; $\dot{m}_{evp}$ is the evaporation rate of fabric, kg/m^3/s; D_l is the liquid diffusion coefficient, m^2/s; ρ_l is the density of liquid water, kg/m^3; T is temperature, °C; ρ_{com} is the effective density of fabric with water, kg/m^3; $c_{v,com}$ is the effective heat capacity of fabric with moisture, J/m^3 K; λ_{com} is the effective thermal conductivity of fabric, W/m/K; Q is the heat source, which is related to water evaporation and heat convection, J.

In the liquid-phase mass transfer, Equation (1), $\dot{m}_{evp}$ is proportional to the difference between the equilibrium vapor pressure (P^*) in the fabric and the current vapor pressure (P_E) at the head space of the fabric (see Equation (4)). The equilibrium vapor pressure (P^*) is related to temperature (T), and it can be calculated using Antoine's equation (Equation (5)), where A, B and C are constants that are determined by the substance (it is water in this work) and temperature[31]

$$\begin{cases} \dot{m}_{evp} = \dfrac{k_{evp}\rho_l(P^* - P_E)}{P_E}, \text{ if } \varepsilon_l > 0 \\ \dot{m}_{evp} = 0, \text{ if } \varepsilon_l < 0 \text{ or } P^* \le P_E \end{cases} \tag{4}$$

where $k_{evp}\,(s^{-1})$ is an evaporation rate constant

$$P^* = 10^{A-B/(C+T)} \tag{5}$$

In the energy balance equation (Equation (2)), the heat source (Q) includes the heat of water evaporation (Q_1) and the heat of convection (Q_2). The heat source of water evaporation is calculated from Equation (6)

$$Q_1 = -\dot{m}_{evp} H_{evp} \tag{6}$$

To solve the governing equations, the effective density and heat capacity of fabric with water are calculated from the phase volume fractions and the pure components according to Equations (7) and (8), respectively. The effective thermal conductivity of fabric with water is calculated from the thermal conductivity of fully water-saturated and dry fabric using Equation (9)[31]

$$\rho_{com} = \varepsilon_l \rho_l + \varepsilon_f \rho_f + \varepsilon_a \rho_a \tag{7}$$

$$c_{v,com} = \frac{(\varepsilon_l \rho_l c_{vl} + \varepsilon_f \rho_f c_{vf} + \varepsilon_a \rho_a c_{va})}{\rho_{com}} \tag{8}$$

$$\lambda_{com} = \lambda_{dry} + \frac{\varepsilon_l}{1-\varepsilon_f}\left(\lambda_{wet} - \lambda_{dry}\right) \tag{9}$$

where ρ_l, ρ_f and ρ_a are the density of water, fiber and air, respectively; kg/m^3; c_{vl}, c_{vf} and c_{va} are the heat capacity of water, fiber and air, respectively, J/m$^3 \cdot$K; k_{wet} and k_{dry} are the thermal conductivity of fully water-saturated and dry fabric, respectively, W/m/K. Considering the high porosity of fabric, the thermal conductivity of fully water-saturated fabric is approximately equal to the thermal conductivity of liquid water in this work.

In porous media, the transport of liquid water through fabric can be approximated as a diffusion process. Therefore, the liquid migrates from the area with a high volume fraction to the area with a low volume fraction when there is a gradient in the volume of liquid. It is known that water movement is induced by the capillary pressure gradient, and the mass flux is described by Darcy's law[32]

$$J = -\frac{k}{\mu}\frac{\Delta P}{L} \tag{10}$$

where J is the mass flux, m^3/s; ΔP is the capillary pressure-difference, Pa; μ is the hydrodynamic viscosity, Pa $\cdot$ s; k is the permeability of a porous medium, m^2; L is the length along the flow direction, m.

The water diffusivity is an unknown function. It combines the relative permeability and derivative of capillary pressure in regard to water content according to Darcy's law (see Equation (11)). Even though various equations for the calculation of relative permeability and solvent capillary pressure in an unsaturated porous medium have been introduced, some of their parameters have to be obtained from experimental data[31]

$$D_l(\varepsilon_l) = -\frac{k(\varepsilon_l)}{\mu}\frac{\partial P(\varepsilon_l)}{\partial \varepsilon_l} \tag{11}$$

In this work, to simplify the calculation, the liquid diffusion coefficient is calculated with

$$\begin{cases} D_l = \alpha\ (\varepsilon_l - \varepsilon_l^*), & \varepsilon_l \geq \varepsilon_l^* \\ D_l = 0, & \varepsilon_l < \varepsilon_l^* \end{cases} \tag{12}$$

where α is a proportionality constant, and it has to be determined by experimental data. Here, ε_l^* is the residual saturation. In this work, we defined α to be 1×10^{-6} m^2/s and ε_l* to be 0.05 on the basis of previous work.[31,33,34]

Initial and boundary conditions

To generate a solution for the governing equations, the initial and boundary conditions should be specified in terms of temperature, humidity and water content. Initially, the fabric is equilibrated to a standard atmosphere of temperature (20 ± 2°C) and humidity (65 ± 2%).

Initial conditions.

$$\begin{aligned} \varepsilon_l(t=0) &= \varepsilon_{l0} \\ T(t=0) &= T_0 \end{aligned} \tag{13}$$

Boundary conditions. According to the heating test of the hygroscopic and cooling function of fabrics,[3] one side of the fabric is exposed to the standard environment, while the other side is in contact with the testing board, and the testing board is heated with a set power.

At $x = 0$, there is a boundary heat source Q_2 generated by convection. $Q_2 = Q_0$.

At $x = L$, we have two boundary conditions to take into account, forced convection heat transfer and heat loss of moisture evaporation

$$Q_1(x=L) = -\frac{k_{evp}\rho_l(P^* - P_E)}{P_E}H_{evp} \tag{14}$$

$$\lambda_{com}\frac{\partial T}{\partial x}\Big|_{x=L} = -\alpha_t(T - T_E) \tag{15}$$

where α_t is the forced convection heat transfer coefficient (W/m^2/K); T is the temperature of the fabric surface that contacts with air, °C; T_E is the temperature of the environment, °C.

Experimental method

The fabric thickness and weight were tested according to the ISO 3801-1977 and EN ISO 5084-2002 standards, respectively. The relative porosity values of fabrics were calculated by Equation (16)[1]

$$P_{por} = \left(1 - \frac{m}{\rho \times e_r}\right) \times 100\% \tag{16}$$

where P_{por} is the relative porosity; m is the fabric weight (kg); ρ is the fiber density (kg/m^3); e_r is the fabric thickness (m).

The heat conductivity of the fabric was calculated by Equation (17)[35]

$$\lambda_{dry} = x(\varepsilon_f\lambda_f + \varepsilon_a\lambda_a) + \frac{y\lambda_f\lambda_a}{\varepsilon_a\lambda_f + \varepsilon_f\lambda_a} \tag{17}$$

where x and y are the effective fractions of parallel and perpendicular fibers for the heat transfer, and $x + y = 1$. In this work, the fabric is regarded as a stationary and homogeneous porous solid. Therefore, we assume that $x = y = 0.5$.

The model output should be compared with lab test results of heating properties to verify the simulation accuracy. Heating properties also were tested using a YG606 II thermal resistance tester, but the test process was different from that of thermal resistance. Briefly, a program controlling heating power was added to the tester, and the temperature was recorded with the heating time. Before conducting the experimental trial, fabric samples were preprocessed. For the case of dry fabric, it was well conditioned for 48 hours at a temperature of 20 ± 2°C and relative humidity of 65 ± 2%. Regarding wet fabric, it was pre-wetted and soaked to have a constant weight of water homogeneously through three steps, as shown in Figure 1. Firstly, the fabric was dipped in distilled water for 5 minutes and passed through a padding mangle until no water dripped from it. Finally, the wet fabric was conditioned at a temperature of 20 ± 2°C and relative humidity of 65 ± 2% until it had the set weight (20 g) of water.

The volume fraction of fiber, liquid water and air was calculated according to the weight of dry fabric and the water weight through Equations (18)–(20)

$$\varepsilon_f = \frac{m_{dry}}{\rho_f Ah} \times 100\% \tag{18}$$

$$\varepsilon_l = \frac{m_l}{\rho_l Ah} \times 100\% \tag{19}$$

$$\varepsilon_a = 1 - \varepsilon_f - \varepsilon_l \tag{20}$$

where m_{dry} is the weight of the dry fabric sample, kg; A is the area of the sample, m^2; m_l is the weight of water in the fabric.

The initial volume fraction of liquid water was calculated according to the constant water weight (m_0) through Equation (21)

$$\varepsilon_{l0} = \frac{m_0}{\rho_l Ah} \times 100\% \tag{21}$$

Results and discussion

Computed results

Considering the widespread use of polyester and nylon fibers in knitted textiles, we implemented the simulation using COMSOL Multiphysics® modeling software for a polyester fabric and a nylon fabric. The specifications of the polyester and nylon fabrics are shown in Table 1. In the numerical computation, the initial condition of temperature is $T_0 = 25$°C. The surrounding conditions were fixed at a temperature of 20°C and relative humidity of 65%. In this case, the wind speed is set at 1 m/s, which is defined as external forced convection in the simulation process. The forced convection heat transfer coefficient α_t is determined by the original computational program of the COMSOL Multiphysics® software. Thermal and mass transfer occur at the external boundary of the fabric. The lower surface is heated under a constant heating power (275 W/m^2). Considering the fabric is not

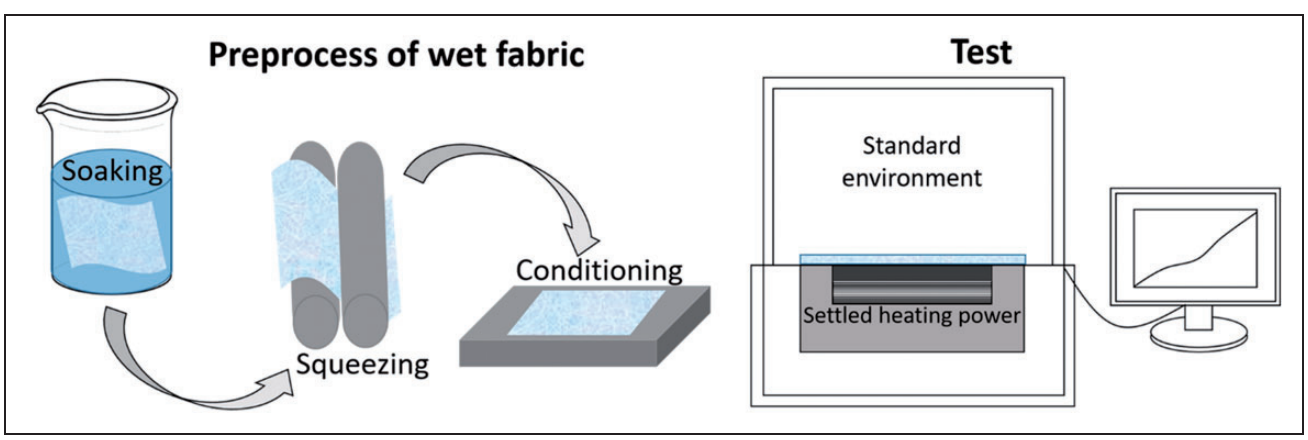

Figure 1. Preprocess of the wet fabric and test method.

Table 1. Specifications of polyester and nylon fabrics

Component	Structure	Wale density (/5cm)	Course density (/5cm)	Weight (g/m^2)	Porosity (%)	Thickness (mm)	Thermal conductivity (10^{-3} W/m/K)
Polyester	Interlock	70	74	196	85.94	1.01	31.48
Nylon	Interlock	68	72	218	78.46	0.88	52.58

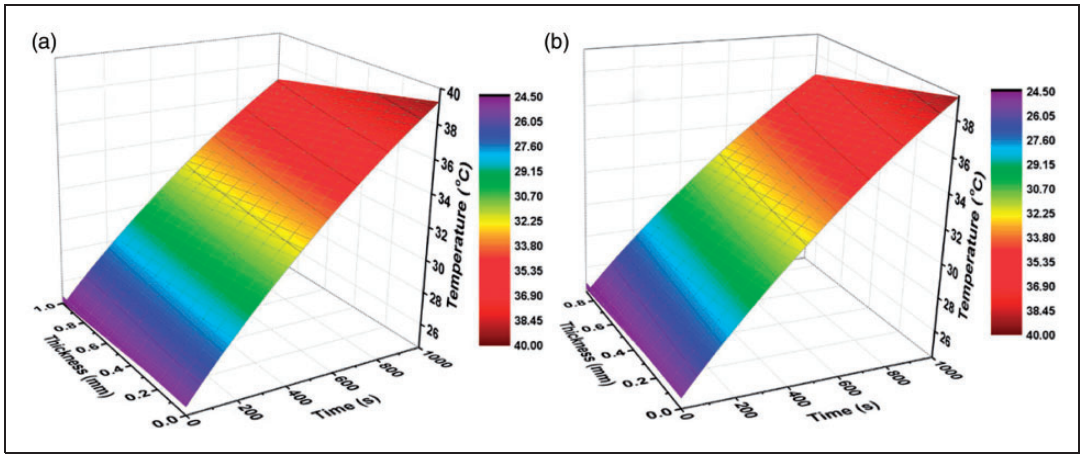

Figure 2. The temperature with time from simulation results of dry polyester (a) and nylon (b) fabrics.

thick, and the water evaporation at the interface of the fabric and air, the evaporative heat loss occurs in the fabric and air at the same time. Hence, the heat lost in the testing system is measured by the water evaporated: $Q_1 = -0.5\dot{m}_{evp}H_{evp}$.

Figure 2 illustrates that the rates of temperature increase with heating time gradually decrease from the simulation results of dry polyester and nylon fabrics. It is known that heat transfer is mainly triggered by the temperature gap,[36] and increasing the temperature gap benefits thermal transfer from a high temperature region to a low temperature region. Thus, with the increase of the temperature gap between the fabric surface and environment, the temperature of the fabric increases faster. The temperature varies slightly through the thickness of the fabric, where a lower temperature is exhibited by the outside surface due to thermal resistance of the fabric.

Figures 3 and 4 demonstrate the simulated temperature and liquid water volume fraction of wet polyester and nylon fabrics, respectively. From Figure 3, it can be seen that there are two stages during the heating process of polyester and nylon fabrics, and they are different from the simulated temperature of dry samples (Figure 1). For instance, the temperature heats up slowly in the first stage, while it increases rapidly in the second stage. In the first stage, liquid water stays within the fabric, and much heat loss is caused by moisture evaporation. In addition, higher thermal conductivity of wet fabric promotes heat transfer, and therefore inhibits the increase of temperature as well. In the second stage, as the heat loss is continuing to decrease, the stored water disappears completely from the fabric, resulting in a rapid increase of temperature. However, there is an obvious difference in these two stages between the polyester and nylon samples due to the variations of thermal and moisture transfer capacities of these two samples. In the heating time of 2000 s, the first stage of the polyester sample is in the range from 0 s to around 1500 s, but the first stage of the nylon sample stays within 0 s to around 1300 s. According to the heating rates, it can be seen that the time at the inflection point of the temperature–time line shows the drying time of wet fabric during the heating process. Thus, the drying time of the nylon sample is lower than that of the polyester sample.

Figure 4 shows that volume fractions of liquid water of these two fabrics decrease at a constant speed at the beginning. This is because vapor pressure on the fabric surface is saturated when there is much liquid water in the fabric. However, when the water content in the fabric is not enough to generate saturated vapor pressure, the change of liquid water volume fraction presents a curvilinear trend until it reaches zero.

Comparison with experiments

To verify this computational model, experiments were carried out by using these two fabrics under identical

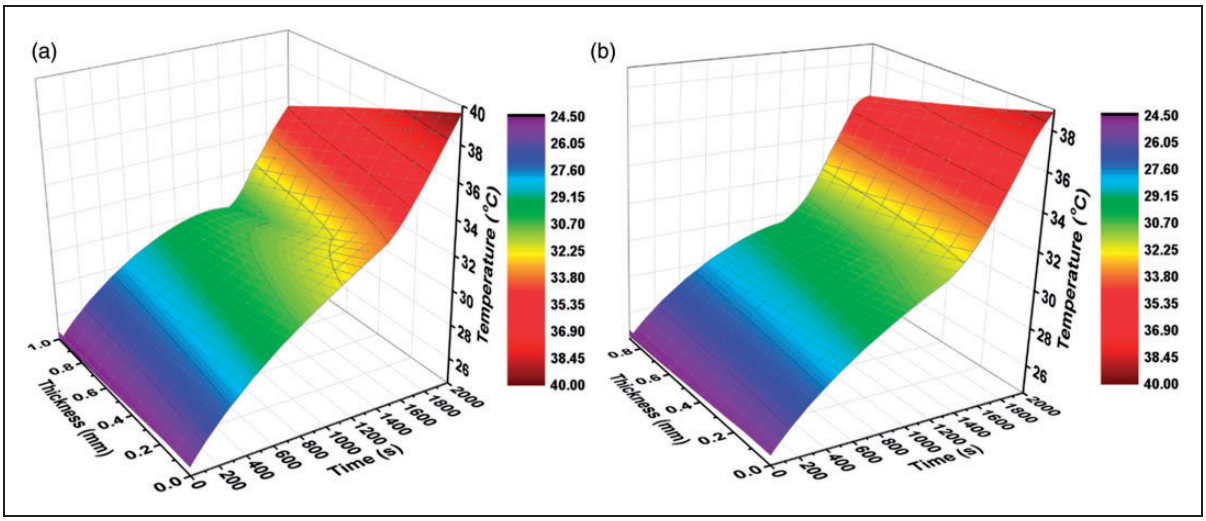

Figure 3. The temperature with time from simulation results of wet polyester (a) and nylon (b) fabrics.

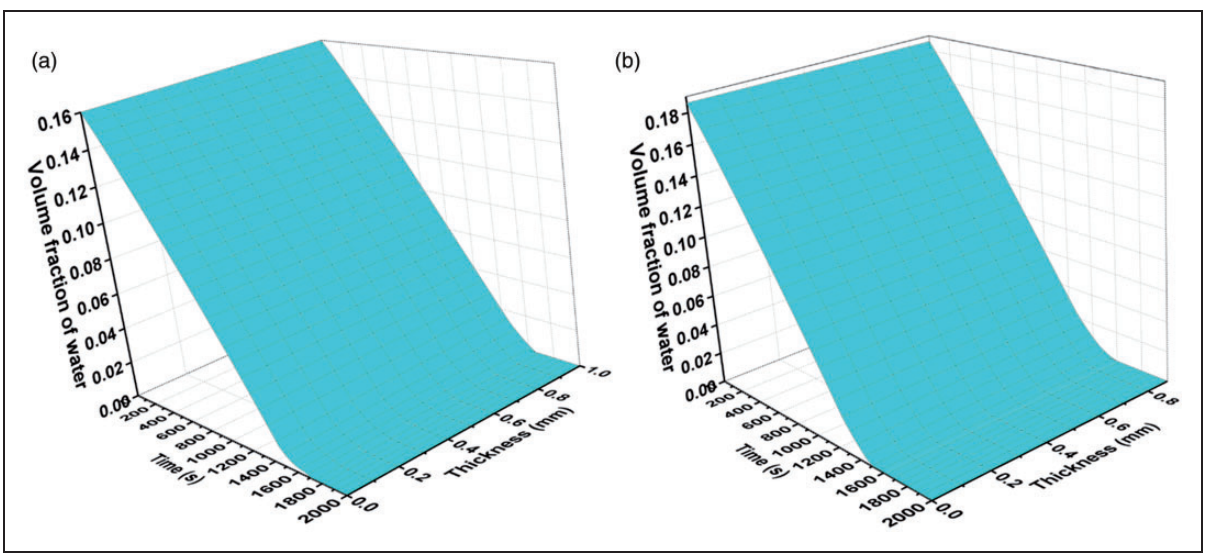

Figure 4. The volume fraction with time from simulation results of wet polyester (a) and nylon (b) fabrics.

conditions. Variations of temperature over heating time at the lower surface of dry and wet fabrics were measured during the heating process. Comparisons of computational temperature variations at the lower surface with experimental results for dry and wet fabrics are shown in Figures 5 and 6, respectively. It can be seen that there are small differences between the computational and experimental curves of polyester and nylon fabrics over heating time due to the proposed assumptions in the model. Figure 7 shows the correlation between the computational and experimental temperature with time of the polyester and nylon fabrics. It is known that this model when used for wet and dry polyester fabrics has R^2 values of 0.9927 and 0.9925, and wet and dry nylon fabrics have R^2 values of 0.9938 and 0.9922, respectively. This demonstrates that the computational results agreed well with the experimental findings for both dry and wet fabrics, indicating that the model can be used to predict the temperature variations at the lower surface of fabric (inner surface) during the heating process for both dry and wet conditions. In our previous work,[3] it was confirmed that the measurement of heating properties of dry and wet fabric was an effective way to evaluate the cooling performance of fabric. Therefore, the evaluation of the cooling performance of fabric can be investigated by this model. From Figure 5, it was shown that the temperature of dry fabric increases with a decreasing rate, whereas the variation of the temperature of wet fabric from 25°C to about 38°C shows a slow increase of

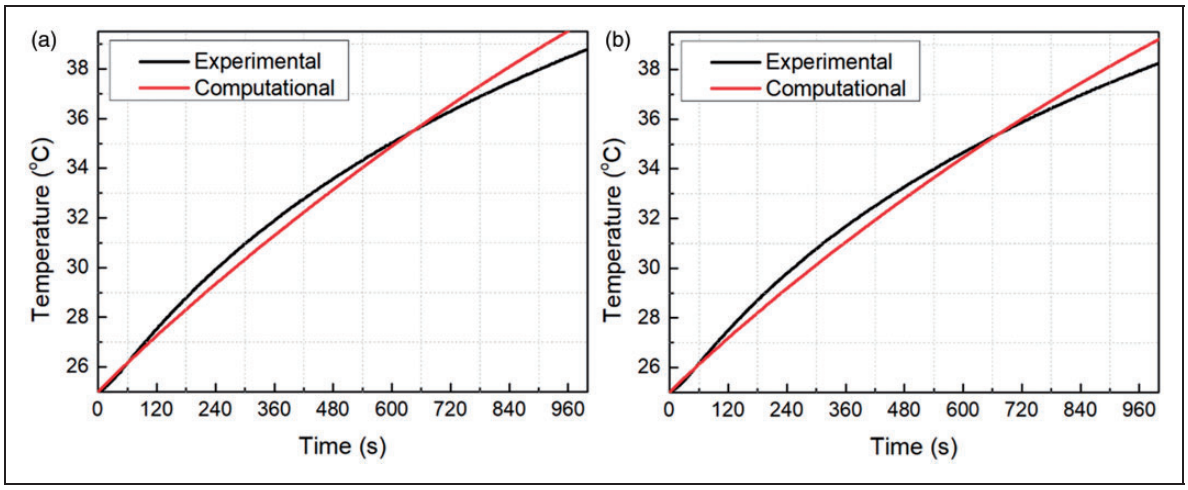

Figure 5. Comparison between computational and experimental temperature changes of dry fabrics: (a) polyester and (b) nylon.

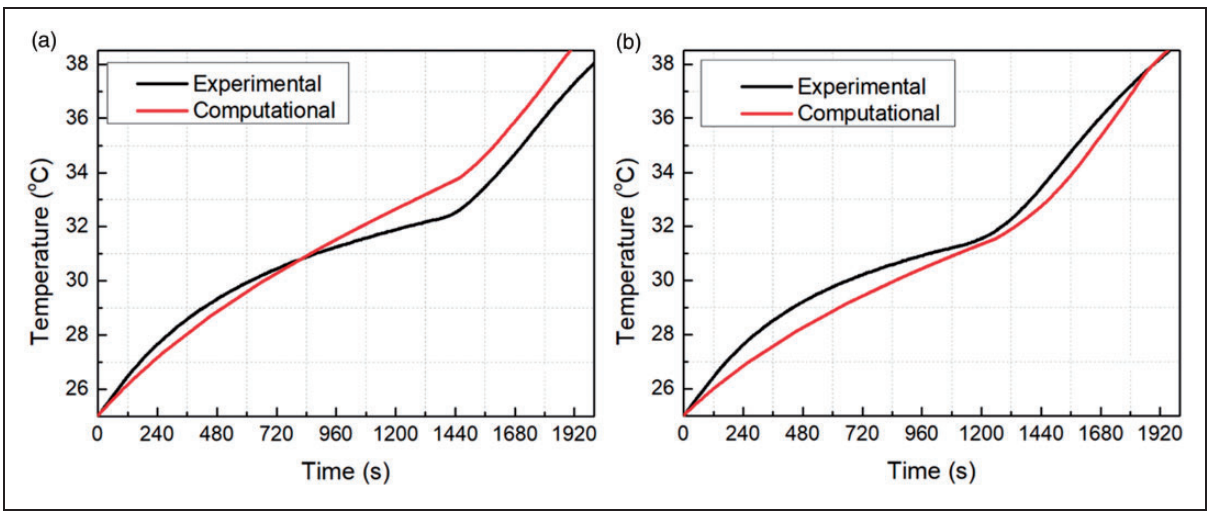

Figure 6. Comparison between computational and experimental temperature changes of wet fabrics: (a) polyester and (b) nylon.

temperature in the initial stage but a significant rise in the second stage due to the variable water content and evaporation rate at the fabric and air interface, as shown in Figure 6.

Influence factors

From the governing equations, it can be shown that the evaporation rate, effective density, thermal conductivity and heat capacity of fabric are important parameters in the computational process, whereas the effective density, thermal conductivity and heat capacity of fabric are dependent on the volume fraction of liquid water in the fabric. Meanwhile, the thermal conductivity of dry fabric plays an important role in the determination of effective thermal conductivity of wet fabric. Therefore, the effects of evaporation rate, volume fraction of liquid water and thermal conductivity of dry fabric on the heating property of fabric are investigated and discussed below, which provides useful information for developing cooling fabrics. The following studies were carried out based on polyester fabric.

Evaporation rate of fabric. In this work, the evaporation rate is determined by k_{evp} and P^*. Here, P^* is a parameter that is related to local temperature. To observe the effect of the evaporation rate of fabric on its heating property, we changed the value of k_{evp} in the computational work. Figure 8 shows a contour map of the temperature changes of fabric with different evaporation rates. It is known that temperature varies along the fabric thickness due to thermal resistance of wet fabric and changes of water content within the fabric. The evaporation rate has significant effect on the

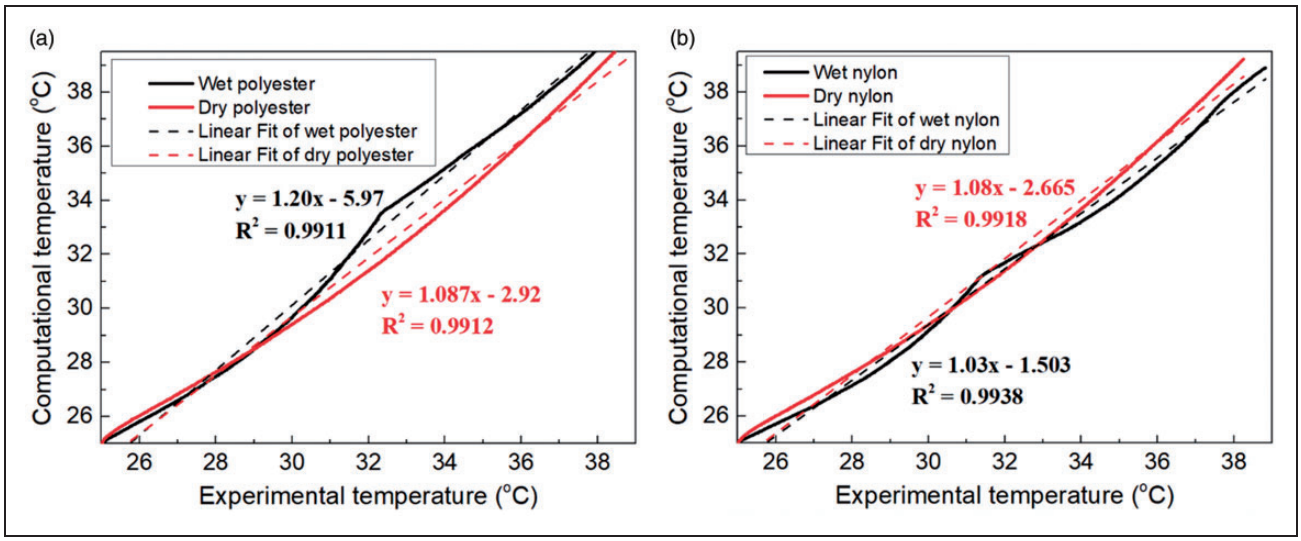

Figure 7. Correlation between computational and experimental temperature of fabrics: (a) polyester and (b) nylon.

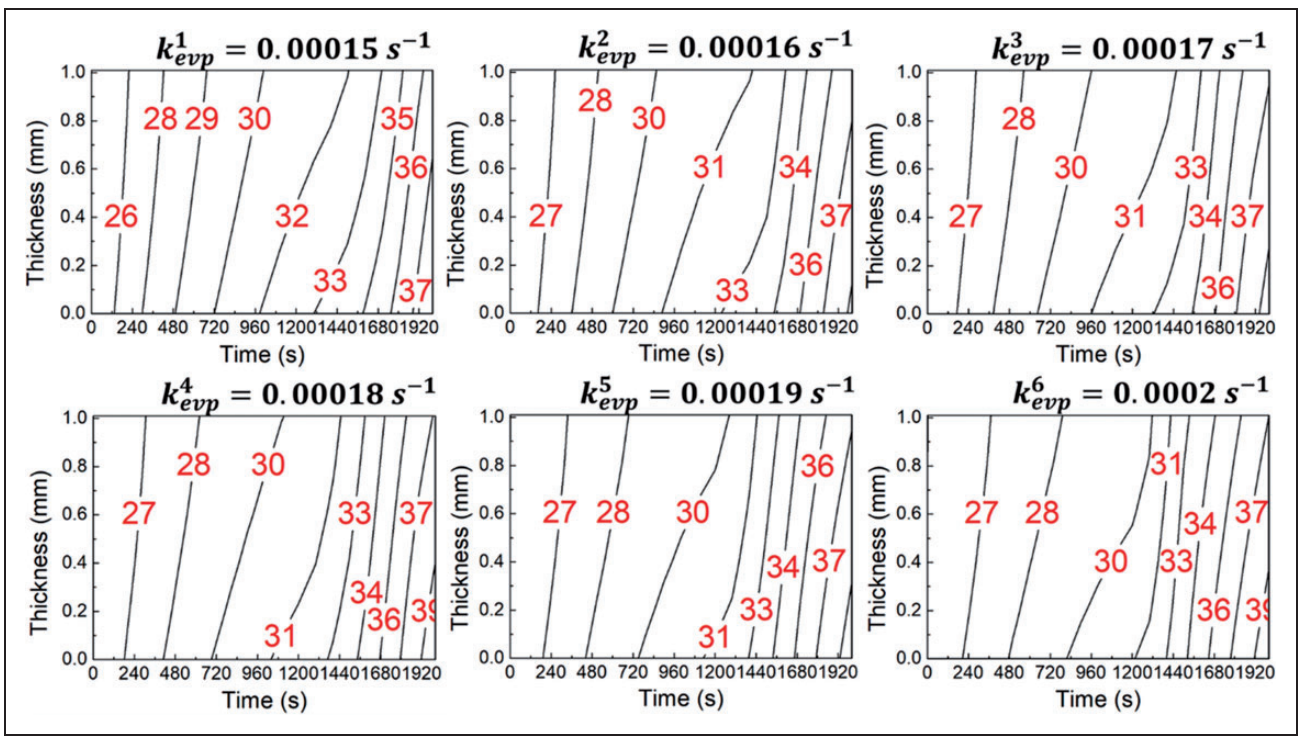

Figure 8. Contour map of temperature changes of fabric with different evaporation rates.

heating property of fabric. As shown in Figure 9(a), with the increase of evaporation rate, the temperature is lower at the first stage and the drying time is shorter, because a quantity of heat was taken away by moisture evaporation and diffusion. Meanwhile, the volume fraction of liquid water decreases fleetingly with increasing evaporation rate (Figure 9(b)). This illustrates that improving the drying performance of fabric is an effective way to enhance its cooling capability and keep the body–clothing microclimate comfortable. This is also in good agreement with the fact that sports clothing often requires the fabric to have an excellent fast-dry performance.

Volume fraction of liquid water. The volume fraction of liquid water in fabric depends on the amount of sweating, which dominates the cooling property of the fabric. As shown in Figure 10, there are obvious

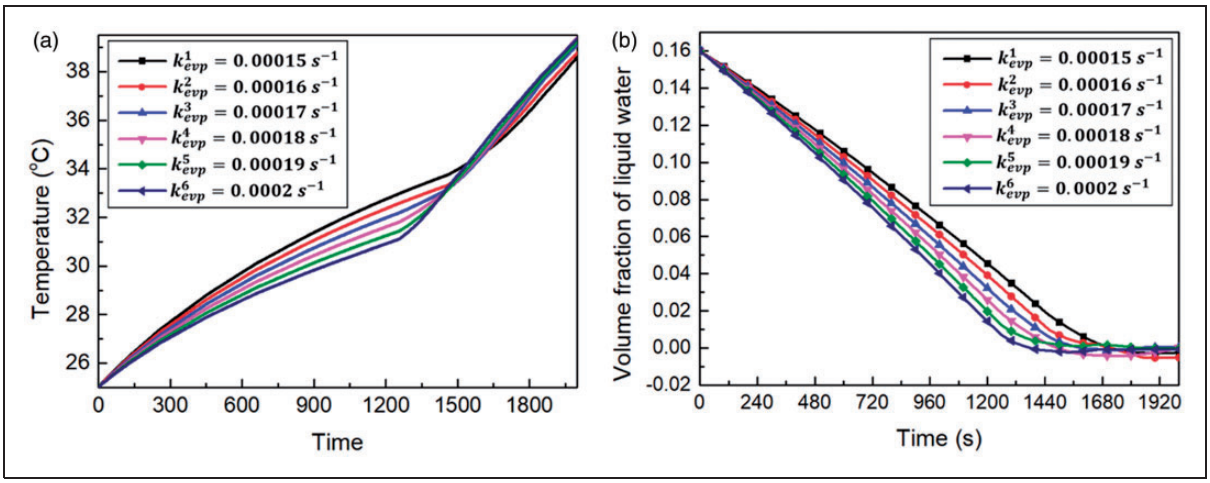

Figure 9. The effects of the evaporation rate of fabrics on temperature changes (a) and volume fraction of liquid water and of its inner surface in the heating process (computed results).

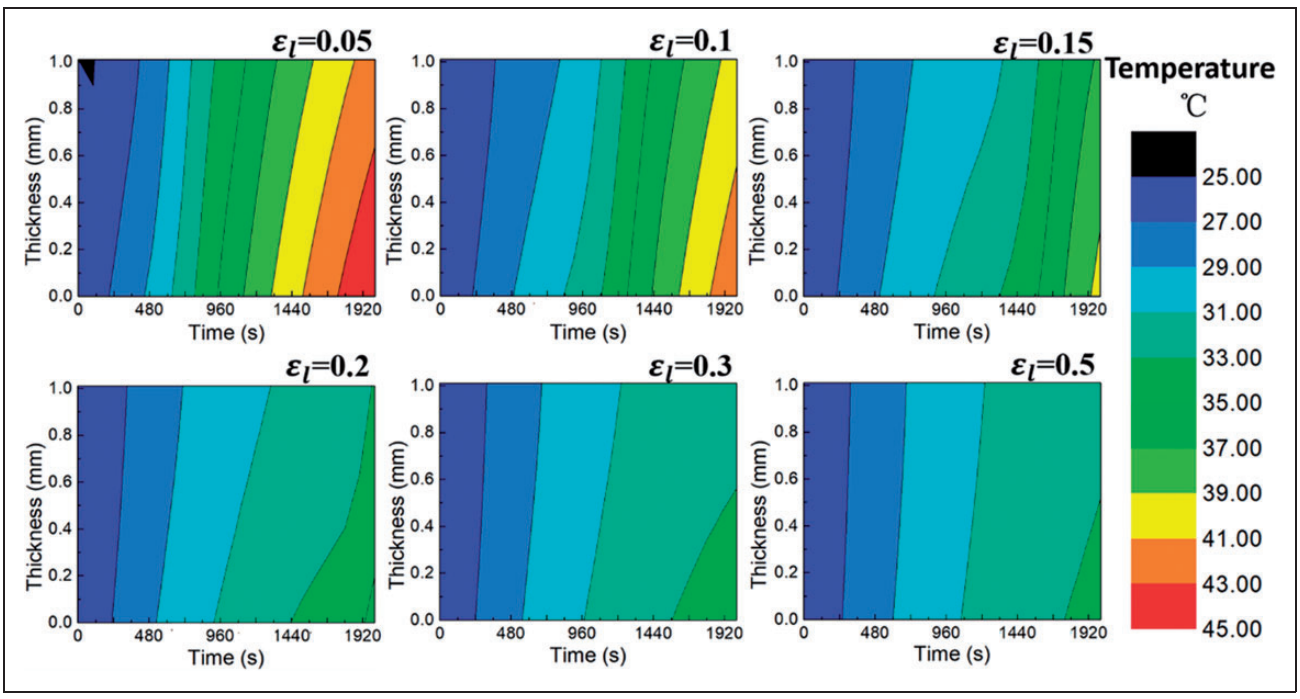

Figure 10. Contour map of temperature changes of fabric with different water content.

temperature changes along the thickness of fabric with varied water content. Figure 11 shows temperature changes (Figure 11(a)) and water content (Figure 11 (b)) at the inner surface of the fabric with varied volume fractions of liquid water from the simulation results. It is observed that there is a lower temperature and longer drying time with an increasing volume fraction of liquid water in the fabric (Figure 11(a)). This illustrates that the evaporation of sweat enhances the cooling performance of fabric during exercise. At the beginning of the heating process, fabrics have similar temperature changes with heating time even though they have varied water content. Meanwhile, the decreasing rate of the volume fraction of liquid water is similar (Figure 11(b)). The relative humidity of the air surrounding fibers is considered to be $RH = 100\%$, and water vapor pressure is saturated when fibers are covered with liquid water. Furthermore, the evaporation rate of fabric depends on the equilibrium vapor pressure, which is determined by temperature according to Equations (4) and (5). Therefore, the temperature difference of fabric with varied water content is

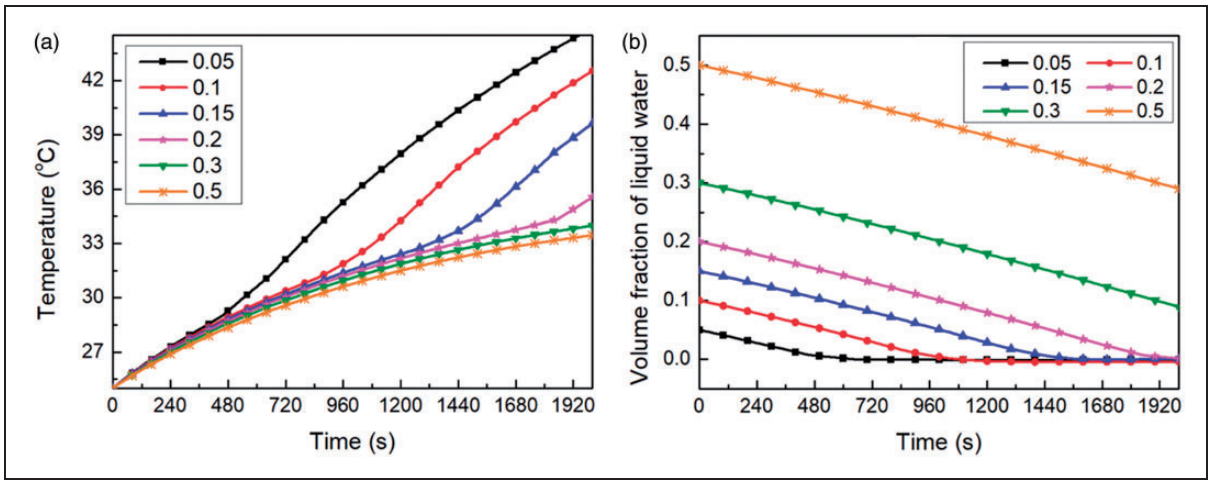

Figure 11. Temperature changes (a) and volume fraction of liquid water changes and (b) of wet fabric with different water content in the heating process (computed results).

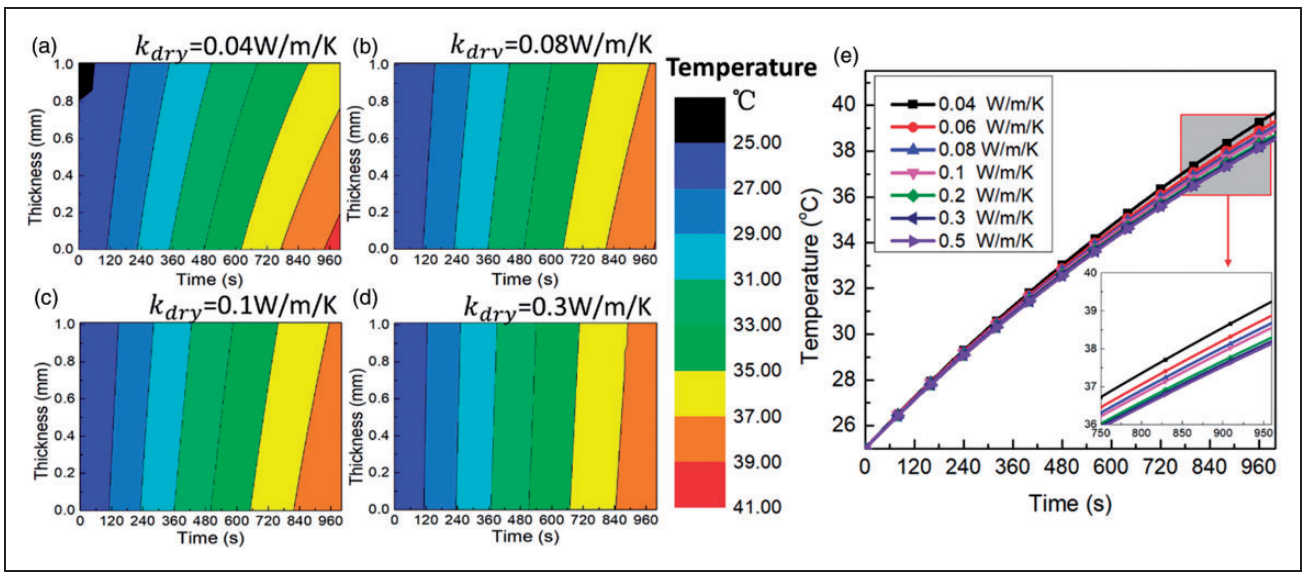

Figure 12. Temperature changes of dry polyester fabrics with different thermal conductivities in the heating process (computed results).

minor. However, there is significant decrease of temperature when the volume fraction of liquid water is more than 0.2 within 2000 s of heating time. This is mainly because more time is needed to evaporate water as the amount of liquid water increases (Figure 11(b)).

Thermal conductivity of fabric. The thermal transfer capacity plays an important role in the thermal comfort property of fabrics. As shown in Figure 12, temperature decreases slightly with the increase of thermal conductivity of dry fabrics. However, the temperature distribution along the fabric thickness changes obviously, as shown in Figures 12(a)–(d). The isothermal lines change from inclined to vertical. Meanwhile, temperature changes of wet fabrics demonstrate similar trends with increasing thermal conductivity (Figure 13). This is because a higher thermal conductivity of fabric is conducive to heat transfer from the inner side to the outer side, which results in a higher temperature at the inner side of fabric in the heating process. In addition, the temperature gap between fabrics decreases with increasing thermal conductivity (Figure 12). For example, the temperature gap between fabrics with thermal conductivities of 0.04 and 0.06 W/m/K is obviously higher than the temperature gap between fabrics with thermal conductivities of 0.06 and 0.08 W/m/K. This is mainly caused by the lower resistance of the thermal transfer of fabric with higher

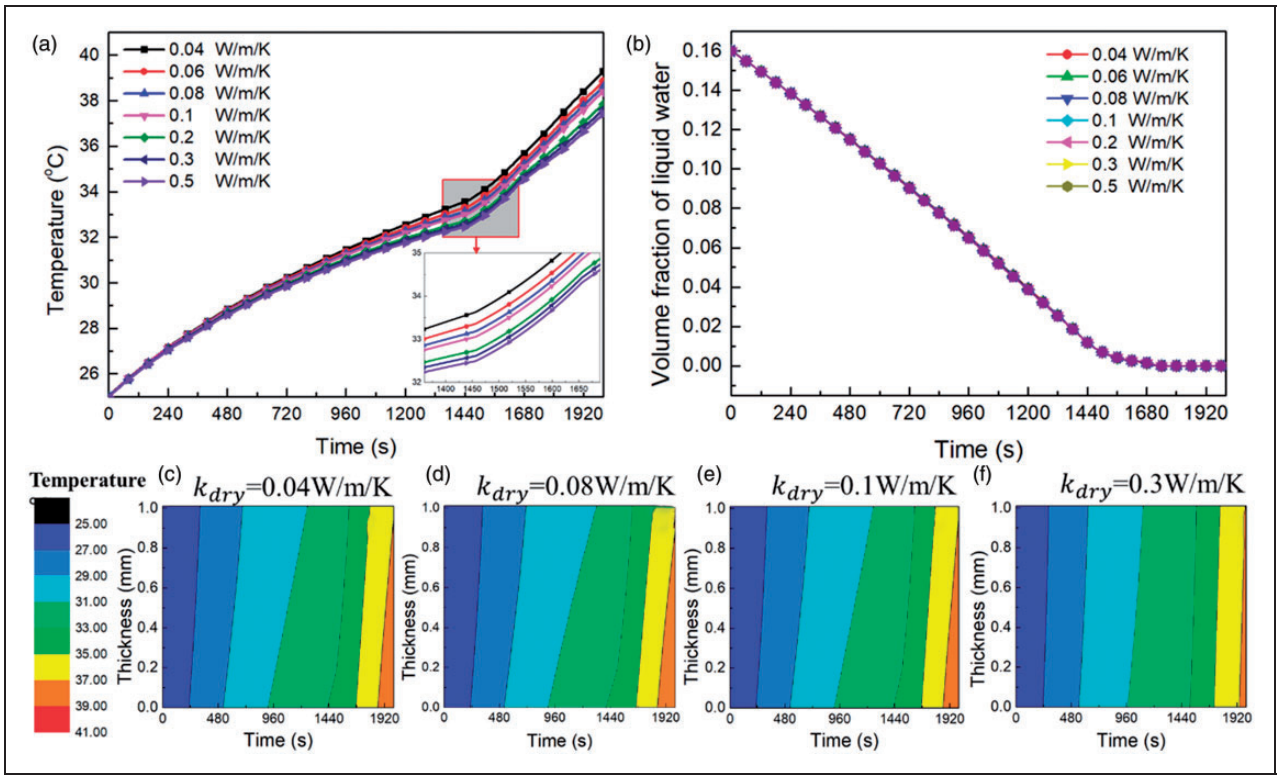

Figure 13. Temperature changes (a) and volume fraction of liquid water changes (b) of wet fabrics with different thermal conductivities in the heating process and (c)–(f) Contour maps of temperature changes (computed results).

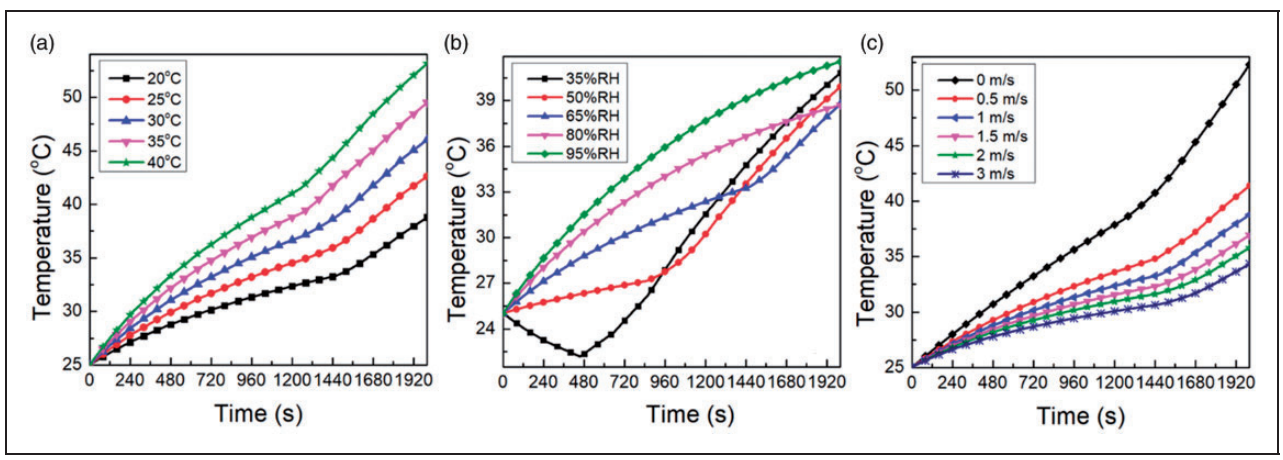

Figure 14. Temperature changes of polyester fabric surrounded by varied conditions in the heating process: (a) temperature; (b) relative humidity and (c) air velocity.

thermal conductivity. From Figure 13(b), it can be seen that thermal conductivity of dry fabrics hardly affects the drying performance of fabrics. Thus, higher thermal conductivity of materials benefits the development of fabric with a cooling effect, whereas increasing the thermal conductivity of fabric has a minor effect on the cooling performance when the thermal conductivity is greater than a certain value.

Surrounding conditions. The thermal and moisture transfer capabilities of fabric are highly affected by its surrounding conditions, such as temperature, relative humidity and air velocity. It is known that heat transfers from high-temperature areas to low-temperature areas, and thus heat transfer capacity is proportional to the temperature gap along the direction of heat transfer according to Fourier's law.[37] As shown in

Figure 14(a), the fabric temperature decreases as its surrounding temperature decreases. Meanwhile, fabric dries faster with the increase of surrounding temperature, because high temperature promotes the evaporation rate of wet fabric, which can be seen from Equations (4) and (5). Figure 14(b) demonstrates that the relative humidity has a significant impact on the thermal and drying properties of wet fabric. It is well known that water vapor pressure increases with increasing relative humidity.[38] At low relative humidity, the drying time is relatively short, and a strong cooling function is exhibited due to the high evaporation rate triggered by the difference of water vapor pressure between the evaporation interface and the surrounding environment (Equation (4)). The temperature at the inner surface of the fabric decreases to 22.2°C even though it is kept heated from 25°C at 35% relative humidity. With the increase of relative humidity, the drying time increases and the cooling function becomes weaker. From Figure 14(c), it can be seen that with the increase of air velocity and temperature, the drying time decreases, which illustrates that wind promotes the cooling efficacy of wet fabric. This is because wind surrounding the fabric induces the air motion and speeds up the diffusion and evaporation of water. Therefore, a windy environment with lower temperature and low relative humidity is a favorable environment for doing exercise.

Conclusion

In this work, the coupled heat and moisture transfer of porous fabrics in the heating process was investigated. A mathematical model of heat and moisture transfer of fabric was implemented to estimate the cooling performance of fabric with low hygroscopicity. The governing equations were established on the basis of the energy balance and the mass balance of liquid water in this work. Specific initial conditions and boundary conditions were set according to the evaluation method of the cooling performance of fabric. Then a series of computations were carried out to establish the coupled heat and moisture transfer model. A cooling property test was conducted to verify the accuracy of this model, and the computed results agreed well with the experimental results. Therefore, this simulation is able to predict the cooling performance of fabric.

The thermal and moisture transfer capacity and surrounding conditions of fabric have significant effects on the cooling heating property according to the results of the simulation. Increasing the evaporation rate, thermal conductivity and water content in the fabric enhances the cooling capability of the fabric. However, high amounts of liquid water in a fabric provide no obvious improvement to its cooling performance. In addition, the surrounding conditions affect the body thermal comfort. It is better to do exercise in an environment with lower temperature, lower relative humidity and/or higher wind speed. These results provide useful information to develop fabrics used for sports and summer clothing.

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ORCID iDs

Yang Yang https://orcid.org/0000-0002-9107-5168
Zheng Liu https://orcid.org/0000-0003-3698-3646
Lin Lou https://orcid.org/0000-0003-4856-9944
Xin Liu https://orcid.org/0000-0001-9223-1858
Peihua Zhang https://orcid.org/0000-0002-1538-4890

References

1. Bedek G, Salaün F, Martinkovska Z, et al. Evaluation of thermal and moisture management properties on knitted fabrics and comparison with a physiological model in warm conditions. *Appl Ergon* 2011; 42: 792–800.
2. Guo SM, Yu DM, Liu YT, et al. Design and development of cool multifunctional composite high-grade fabrics. *Appl Mech Mater* 2014; 633: 476–479.
3. Yang Y, Zhang WJ and Zhang PH. Evaluation method for the hygroscopic and cooling function of knitted fabrics. *Text Res J* 2019; 89: 5024–5040.
4. Xu D and Cui P. Simultaneous determination of thickness, thermal conductivity and porosity in textile material design. *J Inverse III Pose P* 2016; 24: 59–66.
5. Djongyang N, Tchinda R and Njomo D. Thermal comfort: a review paper. *Renew Sust Energ Rev* 2010; 14: 2626–2640.
6. Holcombe BV and Hoschke BN. Dry heat transfer characteristics of underwear fabrics. *Text Res J* 1983; 53: 368–374.
7. Atasağun HG, Okur A, Psikuta A, et al. The effect of garment combinations on thermal comfort of office clothing. *Text Res J* 2019; 89: 4425–4437.

8. Watkins DA and Slater K. 2—The moisture-vapour permeability of textile fabrics. *J Text I* 1981; 72: 11–18.
9. Zhou H, Wang H, Niu H, et al. One-way water-transport cotton fabrics with enhanced cooling effect. *Adv Mater Interfaces* 2016; 3: 1600283.
10. Hes L and Ursache M. Effect of composition of knitted fabrics on their cooling efficiency at simulated sweating. *Indian J Fibre Text* 2011; 36: 281–284.
11. Boguslawska-Baczek M and Hes L. The effect of moisture on thermal resistance and water vapour permeability of Nomex fabrics. *J Mater Sci Eng* 2011; 3: 358–366.
12. Boguslawska-Baczek M and Hes L. Effective water vapour permeability of wet wool fabric and blended fabrics. *Fibres Text East Eur* 2013; 21: 67–71.
13. Bogusławska-Bączek M and Hes L. Thermophysiological properties of dry and wet functional sportswear made of synthetic fibres. *Tekstilec* 2017; 60: 331–338.
14. Fukazawa T and Havenith G. Differences in comfort perception in relation to local and whole body skin wettedness. *Eur J Appl Physiol* 2009; 106: 15–24.
15. Fu M, Weng W, Chen W, et al. Review on modeling heat transfer and thermoregulatory responses in human body. *J Therm Biol* 2016; 62: 189–200.
16. Xu X and Santee WR. Sweat loss prediction using a multi-model approach. *Int J Biometeorol* 2011; 55: 501–508.
17. Henry PS. Diffusion in absorbing media. *Proc Royal Soc B* 1939; 171: 215–241.
18. David HG and Nordon P. Case studies of coupled heat and moisture diffusion in wool beds. *Text Res J* 1969; 39: 166–172.
19. Li Y and Holcombe BV. A two-stage sorption model of the coupled diffusion of moisture and heat in wool fabrics. *Text Res J* 1992; 62: 211–217.
20. Li Y and Holcombe BV. Mathematical simulation of heat and moisture transfer in a human-clothing-environment system. *Text Res J* 1998; 68: 389–397.
21. Wang Z, Li Y, Kowk YL, et al. Mathematical simulation of the perception of fabric thermal and moisture sensations. *Text Res J* 2002; 72: 327–334.
22. Fontana É, Donca R, Mancusi E, et al. Mathematical modeling and numerical simulation of heat and moisture transfer in a porous textile medium. *J Text I* 2016; 107: 672–682.
23. Xu DH, Meibao BG and Zhang HL. Numerical solution of a dynamic model of heat and moisture transfer in porous fabric under low temperature. *Int J Heat Mass Transf* 2013; 61: 149–157.
24. Zhu FL and Li KJ. Modeling heat and moisture transfer within porous textiles under high temperature gradients. *Adv Mater Res* 2012; 455: 1136–1139.
25. Shen H, Xie K, Shi H, et al. Analysis of heat transfer characteristics in textiles and factors affecting thermal properties by modeling. *Text Res J* 2019; 89: 4681–4690.
26. Zaman M, Liu H, Xiao H, et al. Hydrophilic modification of polyester fabric by applying nanocrystalline cellulose containing surface finish. *Carbohyd Polym* 2013; 91: 560–567.
27. Sampath MB, Prakash C and Kumar MS. Influence of moisture management finish on comfort characteristics of knitted fabrics made from different yarns. *Indian J Fibre Text* 2020; 45: 102–108.
28. Zarrinabadi E, Abghari R, Nazari A, et al. Environmental effects of enhancement of mechanical and hydrophobic properties of polyester fabrics using silica/kaolinite/silver nanocomposite: a facile technique for synthesis and RSM optimization. *EurAsia J BioSci* 2018; 12: 437–450.
29. Jia N, Fraenkel HA and Kagan VA. Effects of moisture conditioning methods on mechanical properties of injection molded nylon 6. *J Reinf Plast Compos* 2004; 23: 729–737.
30. Wang F, Zhou X and Wang S. Development processes and property measurements of moisture absorption and quick dry fabrics. *Fibres Text East Eur* 2009; 17: 46–49.
31. Murru M, Giorgio G, Montomoli S, et al. Model-based scale-up of vacuum contact drying of pharmaceutical compounds. *Chem Eng Sci* 2011; 66: 5045–5054.
32. Rosti ME, Pramanik S, Brandt L, et al. The breakdown of Darcy's law in a soft porous material. *Soft Matter* 2020; 16: 939–944.
33. Fan J, Luo Z and Li Y. Heat and moisture transfer with sorption and condensation in porous clothing assemblies and numerical simulation. *Int J Heat Mass Transf* 2000; 43: 2989–3000.
34. Li Y and Zhu Q. A model of coupled liquid moisture and heat transfer in porous textiles with consideration of gravity. *Numer Heat Tr A Appl* 2003; 43: 501–523.
35. Bogaty H, Hollies NR and Harris M. Some thermal properties of fabrics: Part I: the effect of fiber arrangement. *Text Res J* 1957; 27: 445–449.
36. Liu Y, Ye F, Zhang W, et al. Heat transfer behavior caused by temperature difference in reciprocating sliding contact. *Sci Prog* 2020; 103: 1–13.
37. Yang Y, Chen H, Wang H, et al. Optimal thermal rectification of heterojunctions under Fourier law. *Phys Rev E* 2018; 98: 042131.
38. Shi F, Zhao C, Zhou X, et al. Spatial variations of climate-driven trends of water vapor pressure and relative humidity in Northwest China. *Asia-Pac J Atmos Sci* 2019; 55: 221–231.